A3 High Energy Physics Coursework (C2)

Event Classification with Neural Networks

Hadronic Z^0 Decay Flavour Tagging

Xin Li (xl628@cam.ac.uk)

University of Cambridge

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Introduction

Objective

Classify hadronic $Z^0 \to q\bar{q}$ decays into quark flavours ($b\bar{b}$, $c\bar{c}$, $s\bar{s}$) using simulated e^+e^- collision events at the FCC with the IDEA detector, employing neural networks.

- **Context**: e^+e^- collisions at the Z^0-threshold ($\sqrt{s} \approx 91.2$ GeV). The Z^0 decays into quark-antiquark pairs, which subsequently hadronise.

Data Samples

- `Zbb.root`: $Z^0 \to b\bar{b}$
- `Zcc.root`: $Z^0 \to c\bar{c}$
- `Zss.root`: $Z^0 \to s\bar{s}$

Approach

- Traditional methods rely on hand-crafted features (e.g., cut-based analyses).
- Deep learning enables end-to-end classification directly from detector data.

- **Event Classification**: Map detector features $\mathbf{x} \in \mathcal{X}$ to labels $y \in \{b\bar{b}, c\bar{c}, s\bar{s}\}$.
- **Binary Case**: The Neyman–Pearson lemma defines the optimal discriminator:

$$f^*_{\text{N.P.}}(x) = \frac{p(x \mid y=1)}{p(x \mid y=0)}.$$

- Neural networks trained with MSE/cross-entropy loss approximate the **posterior** $p(y=c \mid x)$. $p(y=1 \mid x)$ and $f^*_{\text{N.P.}}(x)$ are related by a one-to-one, monotonic transformation $\Rightarrow$ the ROC curve is invariant to priors.
- In **multi-class** cases, the posterior becomes sensitive to the class priors $p(y)$. [6]

Architectures

- CNNs for image-like data. [5, 2] GNNs for particle interactions. [3] RNNs for time-ordered measurements or ordered particle lists. [1]
- Transformers for modelling long-range dependencies and complex relationships. [4, 7]

Useful Features for Flavour Tagging

- **Displaced Vertices**: b- and c-quarks form long-lived hadrons.
  - Vertex coordinates (x, y, z), errors $(x_{\text{err}}, y_{\text{err}}, z_{\text{err}})$, mass, χ^2, track count.
  - Distance from primary vertex (PV).
- **Leptons**: Heavy-flavour hadrons from b- and c-quarks decay into lighter states, leading to missing energy and charged leptons.
  - Particle ID (muon, electron, etc.), charge, kinematics.
- **Thrust Axis**: Approximates back-to-back quark directions.
  - Components (T_x, T_y, T_z), angle of particles w.r.t. thrust.
- **Global Features**:
  - Total energy, p_T, multiplicities $(n_{\text{particles}}, n_{\text{vertices}}, n_{\text{SV}}, n_{\text{charged}})$.
  - PV fit quality (χ^2_{PV}), PV track count.

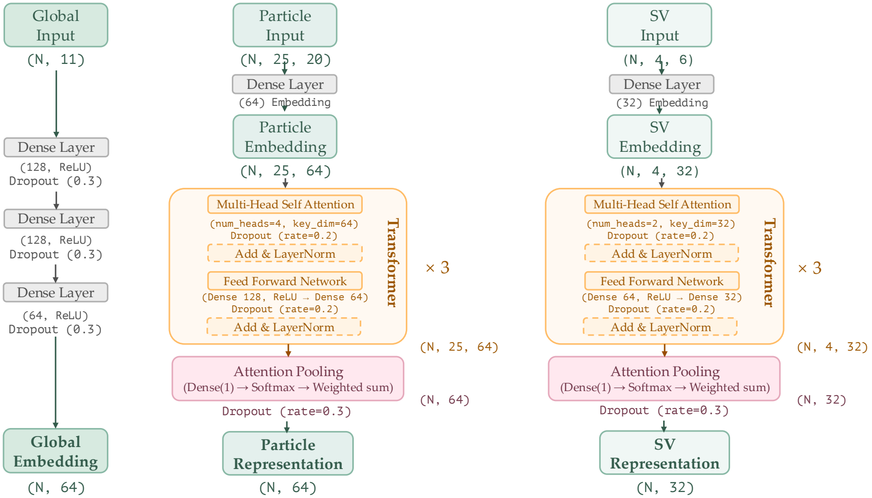
Input Representation

- **Three Feature Groups**:
  - **Global** `(N, 11)`:
    - Thrust components (T_x, T_y, T_z); multiplicities $(n_{\text{particles}}, n_{\text{vertices}}, n_{\text{SV}}, n_{\text{charged}})$; total energy $(E_{\text{tot}}, p_T^{\text{tot}})$; PV fit quality (χ^2_{PV}) and track count $(n_{\text{tracks, PV}})$.
  - **Particles** `(N, 25, 20)`:
    - Momentum components, energy, charge, thrust angle, particle ID (one-hot encoding), charged flag; associated origin vertex features, and displacement from PV.
  - **Secondary Vertices (SVs)** `(N, 4, 6)`:
    - Mass, χ^2, track count, and coordinates (x, y, z).
- Padding and binary masks for particles and SVs: `(N, 25)` $(\mathbf{M}_P)$ and `(N, 4)` $(\mathbf{M}_V)$.

Model Overview

- Transformer-based architecture combining:
  - Global event features `(N, 11)`.
  - Sequences of particles `(N, 25, 20)` and secondary vertices `(N, 4, 6)`.
- Three parallel branches:
  - **Global Branch**: `Dense` layers.
  - **Particle Branch**: `Transformer` layers + attention pooling.
  - **SV Branch**: `Transformer` layers + attention pooling.
- Fusion of all branches via **cross-attention** blocks, followed by concatenation.
- Final prediction produced through dense layers and softmax activation.

Model Architecture: Processing Branches Diagram



Model Architecture: Fusion and Output

- **Cross-Attention**:
  - The global embedding queries the particle and SV embeddings via multi-head attention.
  - Particles and SVs provide the keys and values.
- **Fusion**: Concatenation

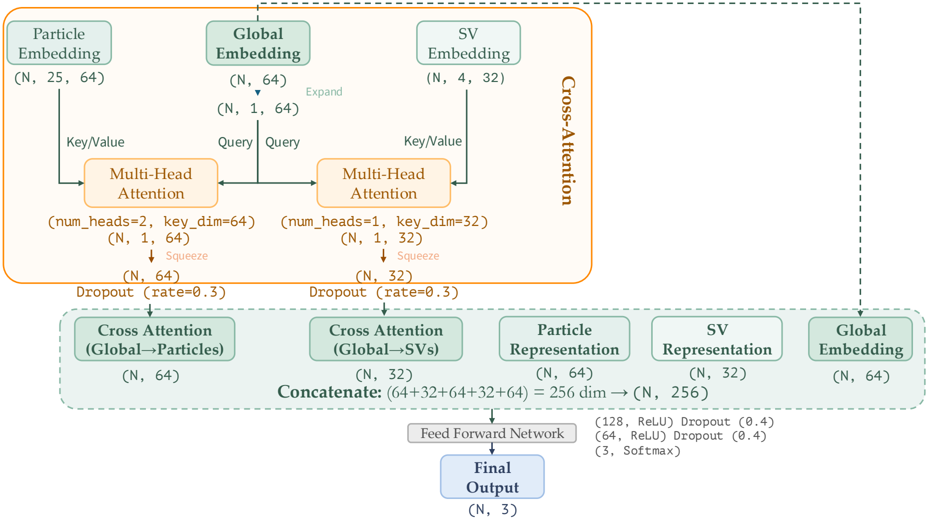
$$\mathbf{z} = [\mathbf{g}', \mathbf{z}_P, \mathbf{z}_V, \mathbf{c}_P, \mathbf{c}_V] \rightarrow \texttt{(N, 256)}$$

where:

  - $\mathbf{g}'$: Global embedding.
  - $\mathbf{z}_P$: Particle self-attention output; $\mathbf{z}_V$: SV self-attention output.
  - $\mathbf{c}_P$: Global-to-particle cross-attention output; $\mathbf{c}_V$: Global-to-SV cross-attention output.
- **Output**: Dense layers followed by softmax:

$$\hat{\mathbf{y}} = \mathrm{softmax}(\mathrm{MLP}_{\mathrm{final}}(\mathbf{z})) \rightarrow \texttt{(N, 3)}$$

Model Architecture: Fusion and Output Diagram



Implementation Details

- **Data Preprocessing**: Apply clipping, padding and masking for variable-length sequences; normalise input features using statistics from the training data.
- **Training**:
  - Epochs: 50; batch size: 256.
  - Learning rate: 0.001, reduced with a patience of 10 epochs.
  - Dropout (0.2–0.4) applied across layers to prevent overfitting.
  - Loss: Cross-entropy for 3-class classification.

Results: Training Dynamics

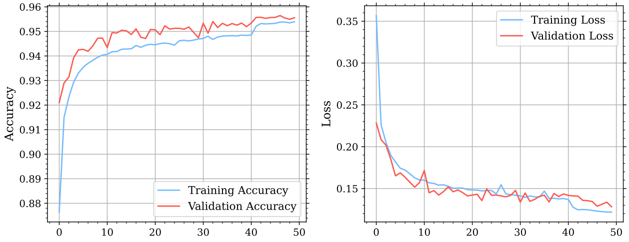
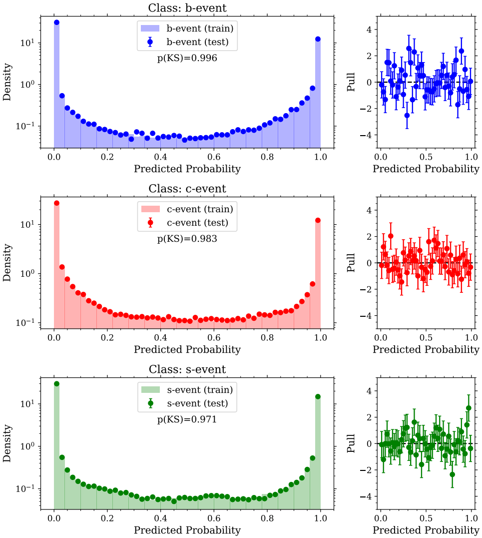


Figure 1: Training/validation accuracy and loss.

- Steady convergence over 50 epochs.
- Learning rate reduction around epoch 40 improved performance.
- No overfitting: Train/validation curves closely aligned.
- Potential for further gains with extended training.

Model Validation: Probabilities and Pull Distributions

- Verify that predicted probabilities are similarly calibrated between training and test sets and that the model does not overfit.
- **Kolmogorov–Smirnov (KS) test** p-values:
  - **b-event**: $p_{KS} = 0.996$
  - **c-event**: $p_{KS} = 0.983$
  - **s-event**: $p_{KS} = 0.971$
- **Observation**: Excellent agreement between training and test distributions across all classes.



Results: Performance Metrics

Table 1: Classification Performance (Test Set)

Class	Precision	Recall	F1-Score	Support
b-event	0.962	0.961	0.962	29,710
c-event	0.934	0.934	0.934	29,968
s-event	0.969	0.970	0.969	31,854
Accuracy		0.955		
Macro Avg	0.955	0.955	0.955	91,532

- **Accuracy**: 95.5% on the test set.
- **Observation**: High precision and recall, balanced across classes; ***c***-events are slightly harder to classify.

Results: Probability Histograms

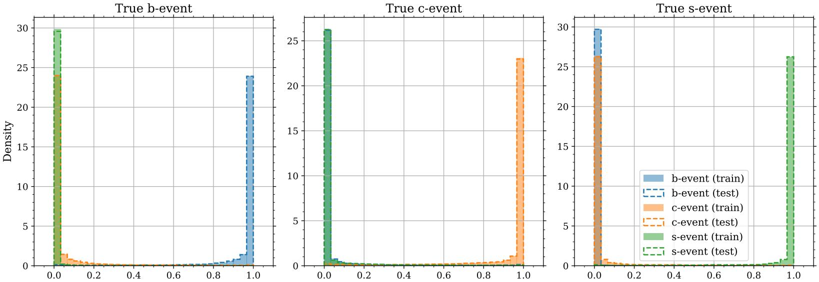


Figure 2: Predicted probability histograms.

- **Discrimination**: Sharp separation between true class probabilities (near **1**) and others (near **0**).
- **Generalisation**: Training and test distributions are nearly identical, indicating robust performance.

Results: ROC and Precision-Recall Curves

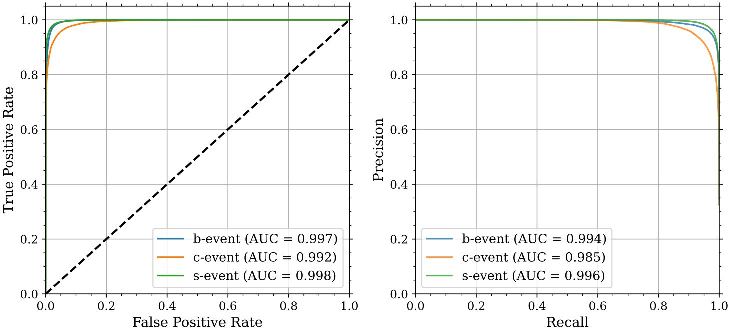


Figure 3: ROC and Precision-Recall curves.

- **ROC/PR**: AUC > 0.99 for nearly all classes, demonstrating excellent sensitivity and precision. Performance remains robust with slight class imbalance.

Conclusion and Future Work

- **Summary**:
  - Transformer model achieves 95.5% test accuracy, with AUC > 0.99.
  - Excels with raw/low-level features, leveraging self- and cross-attention to capture local and global relationships within the data.
  - Demonstrates robust generalisation, with no overfitting observed.
- **Future Work**:
  - Extend training or fine-tune hyperparameters.
  - Explore Graph Neural Networks or DeepSets for modelling particle interactions.
  - Apply to real data or additional decay modes (current data consists purely of three classes).

Thank You! 🤗

Questions?

References I

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[2] A. Aurisano et al. **"A convolutional neural network neutrino event classifier".** In: *Journal of Instrumentation* 11.09 (Sept. 2016), P09001–P09001. ISSN: 1748-0221. DOI: `10.1088/1748-0221/11/09/p09001`. URL: `http://dx.doi.org/10.1088/1748-0221/11/09/P09001`.

[3] A. Aurisano et al. **"Graph neural network for neutrino physics event reconstruction".** In: *Phys. Rev. D* 110 (3 Aug. 2024), p. 032008. DOI: `10.1103/PhysRevD.110.032008`. URL: `https://link.aps.org/doi/10.1103/PhysRevD.110.032008`.

[4] Freya Blekman et al. **“Tagging more quark jet flavours at FCC-ee at 91 GeV with a transformer-based neural network”.** In: *Eur. Phys. J. C* 85.2 (2025), p. 165. DOI: `10.1140/epjc/s10052-025-13785-y`. arXiv: `2406.08590 [hep-ex]`.

[5] S.V. Chekanov. **“Imaging particle collision data for event classification using machine learning”.** In: *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 931 (July 2019), pp. 92–99. ISSN: 0168-9002. DOI: `10.1016/j.nima.2019.04.031`. URL: `http://dx.doi.org/10.1016/j.nima.2019.04.031`.

[6] Particle Data Group. ***Review of Machine Learning in Particle Physics.*** Accessed: 2025-04-06. 2023. URL: `https://pdg.lbl.gov/2023/reviews/rpp2023-rev-machine-learning.pdf`.

[7] Huilin Qu, Congqiao Li and Sitian Qian. *Particle Transformer for Jet Tagging*. 2024. arXiv: 2202.03772 [hep-ph]. URL: https://arxiv.org/abs/2202.03772.